

Legends

Legends can be used to interpret the scales with respect to colour, shapes or sizes. This is similar to axes, where the axes configuration is applied to interpret the spatial ranges. It is mandatory to specify the size, shape, and whether to fill or stroke either one of the parameters for the legend to uniquely differentiate the group.

Marks

A Marks section is the most important in the Vega specification and is the basic building block of visualisation. This section defines the type of plotting to visualise the data such as arc, rectangles, line, area, symbols and images. The special mark type *group* is used to group elements and visualise the data.

```
"marks": [
  {
    "type": "rect",
    "from": {"data": "table"},
    "properties": {
      "enter": {
        "x": {"scale": "x", "field": "data.x"},
        "width": {"scale": "x", "band": true, "offset": -1},
        "y": {"scale": "y", "field": "data.y"},
        "y2": {"scale": "y", "value": 0}
      },
      "update": { "fill": { "value": "steelblue" } },
      "hover": { "fill": { "value": "red" } }
    }
  }
]
```

We have created a Vega visualisation using employment statistics from <http://data.gov.in/> and it is available in <https://github.com/karthikeyan-r/VegaGrammar>.

Vega runtime

Vega provides a runtime that takes Vega specification as input and displays the visualisation in the browser using HTML5 Canvas. The following code snippet shows how this is done:

```
<html>
<head>
<title>Vega Runtime Sample</title>
<!-- Add reference to D3, Vega JS -->
<script src="http://trifacta.github.io/vega/lib/d3.v3.min.js"></script>
<script src="http://trifacta.github.io/vega/lib/d3.geo.projection.min.js"></script>
<script src="http://trifacta.github.io/vega/vega.js"></script>
</head>
```

```
<body>
<div id="vis"></div>
</body>
<script type="text/javascript">
// Parse a vega specification @URL and create a visualization
view and add
// chart component to DIV @vis
// vg.parse.spec is the entry point for the runtime to render
the chart.
function parse(spec) {
  vg.parse.spec(spec, function(chart) { chart({el: "#vis"}).
  update(); });
}
parse("/vegaSpecificationURL.json");
</script>
</html>
```

The callback function takes the name of the DOM element and the set of optional arguments, such as the data to visualise on-demand and some basic properties related to the viewport.

Graphviz

Graphviz is a package of open source tools initiated by AT&T Labs Research for drawing graphs. Graph is a structural representation of data and is composed of vertices and edges. Some of the popular forms of graph diagrams are sociograms, state diagrams, flowcharts and computer network diagrams.

Graphviz takes the description of graphs in a simple text language called DOT and creates graphs in formats like SVG, PDF and Postscript. Graphviz includes a command line tool called dot which creates graphs in various output formats. It also includes multiple layout engines like *circo*, *twopi*, *fdp*, *neato*, etc, and a graphical user interface called *dotty* to visualise and edit graphs. Since we are interested in the DSL aspect of Graphviz, we will focus on DOT and its capabilities. For more information about various utilities bundled with Graphviz, please take a look at the References given at the end of the article.

Graph Description Language—DOT

DOT is a simple but powerful graph description language with C style syntax. A DOT file is just a text file with a .gv or a .dot extension. Various utilities in the Graphviz package take this file as input and produce the graphs. There are even some utilities like *gvpr*, *gc*, *acyclic*, *ccomp*, *sccmap* and *tred*, which take the DOT file as input and perform calculations on the represented graph. Though most of the utilities with Graphviz can be run in the command line, you may use GVEdit for working DOT files. It is a handy editor that provides the bare minimum development environment. Using GVEdit, you can create DOT files, visualise the graphs, switch between various layout engines and save the result in various formats like PDF, PNG, etc.